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(57) Abstract:

ABSTRACT The present invention relates to a computer-implemented system and method for real-time multilingual communication. The system comprises an input interface (100), a speech recognition and translation module (110), a multimodal emotion recognition engine (120), a fusion core (130), an adaptive translation policy engine (140), a privacy-preserving embedding layer (150), a predictive misunderstanding detection module (160), an emotion-preserving speech synthesis module (170), and an expression-aligned visual rendering module (180). The system is configured to receive speech and visual inputs, extract emotion features from multiple modalities, generate a unified emotion embedding, and adapt translation behavior based on the detected emotional context. The system further generates translated speech and corresponding visual output while preserving emotional characteristics and analyzing conversational inputs to detect potential misunderstandings. The invention enables real-time, emotion-aware and context-sensitive cross-lingual communication. Fig 1

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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The Patents Rules, 2003

COMPLETE SPECIFICATION

(see section 10)

A REAL-TIME MULTIMODAL SYSTEM FOR ADAPTIVE EMOTION-AWARE

HINDI-ENGLISH COMMUNICATION

The following specification particularly describes the invention and the manner in which it is to be performed

FIELD OF INVENTION

The present invention relates generally to the field of artificial intelligence-based communication systems, and more particularly to a real-time multimodal adaptive translation system for Hindi–English communication. More specifically, the invention pertains to technologies involving multimodal emotion recognition from speech, facial expressions, and prosodic signals, adaptive machine translation, emotion-preserving speech synthesis, privacy-preserving data processing, and predictive conversational analysis. The invention lies at the intersection of natural language processing, speech processing, affective computing, and deep learning-based multimodal systems, and is directed towards enabling emotion-aware, context-sensitive, and real-time cross-lingual communication.

BACKGROUND OF THE INVENTION

In recent years, multilingual communication systems have witnessed significant advancements with the adoption of artificial intelligence, particularly in the domains of machine translation and speech processing. Systems designed for cross-lingual communication, especially between widely used languages such as Hindi and English, have enabled improved accessibility and interaction across diverse linguistic populations. However, most existing systems primarily focus on linguistic translation and treat communication as a text or speech conversion task, thereby neglecting the emotional and contextual nuances inherent in human communication. As a result, such systems often fail to preserve the speaker’s intent, tone, and affective meaning, leading to misinterpretation, reduced communication quality, and cultural disconnect.

One such system is disclosed in CN101937431A, which describes an emotional speech translation device capable of extracting emotional information from source speech and incorporating it into the translation process. The system converts speech into text, derives emotional features, and uses these features during translation and speech synthesis to generate output that reflects the emotional tone of the input. While this approach improves translation fidelity by including emotional cues, the system primarily relies on speech-based emotion extraction and does not account for other modalities such as facial expressions or prosodic variations that contribute significantly to human emotional perception.

Another relevant prior art is US9117446B2, which discloses a method and system for generating emotionally expressive speech from textual input using emotion tags. In this system, text is segmented into rhythm units, and each unit is assigned an emotion vector comprising scores corresponding to different emotional categories. These emotion vectors guide the text-to-speech synthesis process, enabling generation of speech with controlled emotional characteristics. Although this system enables fine-grained emotional speech synthesis, it operates primarily at the text level and does not incorporate real-time multimodal inputs or adaptive translation mechanisms.

Further, US11227195B2 describes a multimodal sentiment detection system that processes video, audio, and textual data to determine sentiment and demographic characteristics of a subject. The system extracts features from visual frames, acoustic signals, and textual transcripts, and combines them into a unified feature vector for classification using machine learning models. This approach demonstrates the effectiveness of multimodal data fusion for sentiment analysis; however, it is primarily directed towards classification tasks and is not integrated with translation systems or real-time communication frameworks.

Additional prior art identified in the prior-art search report includes systems such as emotion-aware machine translation models, multimodal emotion recognition frameworks, prosody-preserving speech translation systems, privacy-preserving speech processing techniques, and dialogue-based misunderstanding detection systems. These systems address specific aspects of communication such as emotion detection, translation adaptation, or privacy protection. However, each of these approaches operates in isolation and focuses on a limited subset of functionalities without providing a unified framework that integrates multiple modalities, adaptive translation behavior, and conversational intelligence.

A common limitation across these systems is the absence of a comprehensive architecture that simultaneously considers multimodal emotional inputs, dynamically adapts translation strategies based on emotional and contextual cues, preserves emotional characteristics in synthesized speech, ensures privacy of sensitive emotional data, and proactively monitors conversational dynamics to prevent misunderstandings. Existing systems also lack the capability to handle multi-participant scenarios in real-time communication environments, where different speakers may exhibit varying emotional states.

Accordingly, there exists a need for an improved communication system that overcomes these limitations by providing a unified, real-time framework capable of integrating multimodal emotion recognition, adaptive translation policy selection, emotion-preserving speech synthesis, privacy-aware data processing, and predictive conversational analysis. The present invention addresses this need by introducing a system in which emotional context derived from multiple modalities directly influences translation behavior and downstream communication outputs, thereby enabling more accurate, context-aware, and effective cross-lingual communication.

SUMMARY OF THE INVENTION

The following summary is provided to facilitate a clear understanding of the new features in the disclosed embodiment and it is not intended to be a full, detailed description. A detailed description of all the aspects of the disclosed invention can be understood by reviewing the full specification, the drawing and the claims and the abstract, as a whole.

The present invention provides a real-time multimodal communication system configured to enable adaptive and emotion-aware Hindi–English translation by integrating multimodal emotion recognition, adaptive translation control, emotion-preserving speech synthesis, privacy-aware processing, and predictive conversational analysis within a unified architecture.

In an aspect, the invention provides a computer-implemented system for real-time cross-lingual communication, wherein input speech in a source language is processed to generate translated output in a target language while preserving emotional context. The system is configured to receive multimodal inputs including speech signals, facial expressions, and prosodic features, and to generate corresponding translated speech and visual outputs.

In another aspect, the invention provides a multimodal emotion recognition module configured to extract emotion features from speech, facial images, and prosodic signals, and to generate a unified emotion embedding using a cross-modal attention-based fusion mechanism. The fused emotion representation is indicative of an emotional state selected from a plurality of predefined emotion categories.

In another aspect, the invention provides an adaptive translation policy engine configured to determine a translation strategy based on the generated emotion embedding and contextual parameters. The translation strategy comprises one or more control modes including formal,

115 casual, empathetic, simplified, and clarification-oriented translation, thereby enabling dynamic modification of translation behavior.

In another aspect, the invention provides a translation and synthesis module configured to generate translated textual content and corresponding speech output, wherein the speech output is synthesized using emotion-conditioned parameters to preserve prosodic and affective
120 characteristics of the input speech.

In another aspect, the invention provides an expression-aligned rendering module configured to generate a visual representation of a speaker, wherein facial expressions of an avatar are controlled based on the detected emotional state using predefined facial action mappings.

In another aspect, the invention provides a privacy-aware embedding module configured to apply
125 differential privacy to the generated emotion embedding by introducing calibrated noise, thereby enabling protection of sensitive emotional information while maintaining utility for downstream processing.

In another aspect, the invention provides a predictive misunderstanding detection module configured to analyze a sequence of conversational inputs and generate a risk score indicative of a
130 potential communication breakdown, wherein the system is further configured to initiate corrective actions including clarification prompts, simplification of translation, or modulation of communication pace.

In another aspect, the invention provides an integrated processing pipeline wherein the multimodal emotion recognition, translation policy selection, translation, synthesis, rendering, privacy
135 protection, and misunderstanding detection modules operate in coordination to process conversational inputs in real time.

The invention addresses the objective technical problem of enabling accurate and context-aware cross-lingual communication by overcoming limitations of conventional systems that fail to incorporate emotional context, lack adaptive translation mechanisms, and do not provide proactive
140 conversational monitoring. The proposed solution integrates multimodal emotion detection with adaptive translation control and real-time synthesis to preserve both semantic and affective content.

The key inventive features of the invention include a transformer-based cross-modal emotion fusion mechanism, an emotion-conditioned adaptive translation policy engine, an emotion-preserving bilingual speech synthesis framework, a privacy-preserving emotion embedding mechanism, and a predictive misunderstanding detection system integrated within a unified real-time communication pipeline.

The technical advantages of the invention include improved preservation of emotional intent during translation, enhanced communication accuracy, dynamic adaptation of translation behavior based on emotional context, reduced risk of miscommunication through predictive intervention, and improved privacy protection for sensitive emotional data.

Accordingly, the invention provides a technically advanced communication system that enables real-time, emotion-aware, and context-sensitive Hindi–English interaction through an integrated multimodal architecture.

BRIEF DESCRIPTION OF THE DRAWINGS

The manner in which the present invention is formulated is given a more particular description below, briefly summarized above, may be had by reference to the components, some of which is illustrated in the appended drawing. It is to be noted; however, that the appended drawing illustrates only typical embodiments of this invention and are therefore should not be considered limiting of its scope, for the system may admit to other equally effective embodiments.

Throughout the drawings, the same drawing reference numerals will be understood to refer to the same elements and features.

The features and advantages of the present invention will become more apparent from the following detailed description along with the accompanying figures, which forms a part of this application and in which:

Fig. 1 illustrates a system architecture diagram of a real-time multimodal adaptive Hindi–English communication system, depicting a plurality of interconnected modules and data flow within a unified processing pipeline.

Fig. 2 illustrates confusion matrices corresponding to classification models evaluated on test samples, including speech-based emotion classification, vision-based facial emotion classification, multimodal fusion-based emotion classification, and translation policy classification.

Fig. 3 illustrates a comparative performance analysis of multiple models in terms of accuracy and F1-score, highlighting performance improvement achieved through multimodal fusion.

Fig. 4 illustrates modality attention analysis generated by a cross-modal fusion mechanism, showing distribution and variability of attention weights assigned to speech, facial, and prosodic inputs.

Fig. 5 illustrates performance evaluation metrics including accuracy, precision, recall, and F1-score, along with inference time analysis for individual models and overall system latency.

Fig. 6 illustrates analysis of a predictive misunderstanding detection module, including a receiver operating characteristic (ROC) curve, risk score distribution, and intervention probability outputs.

Fig. 7 illustrates a multi-metric radar comparison of different models based on performance parameters including accuracy, precision, recall, F1-score, and computational efficiency.

Fig. 8 illustrates experimental evidence of dynamic cross-modal attention fusion, including modality-wise attention distributions, emotion-conditioned attention patterns, and adaptive weighting across samples.

Fig. 9 illustrates experimental results of an adaptive translation policy engine, including policy selection distribution, emotion-conditioned policy mapping, confidence calibration, and classification accuracy.

Fig. 10 illustrates experimental analysis of predictive misunderstanding detection, including risk distribution, intervention probability mapping, temporal progression of conversational risk, and ROC performance.

Fig. 11 illustrates performance evaluation of an end-to-end real-time processing pipeline, including latency breakdown, throughput analysis, real-time compliance, and model-wise inference comparison.

195 Fig. 12 illustrates experimental validation of a unified hierarchical embedding space, including embedding projections, clustering quality analysis, and hierarchical representation flow.

Fig. 13 illustrates detailed analysis of a translation policy engine, including policy distribution, confidence score variation, and emotion-wise policy selection behavior.

200 Fig. 14 illustrates multi-person facial emotion analysis from video frames using a vision-based detection system, demonstrating detection accuracy, emotion distribution, comparative performance, confidence levels, intensity analysis, and simultaneous analysis of multiple participants.

List of Reference Numerals

- 100 - Input Interface (Microphone and Camera)
- 205 110 - Speech Recognition and Translation Module
- 120 -Multimodal Emotion Recognition Engine
- 120A - Speech Encoder
- 120B - Facial Encoder
- 120C - Prosodic Encoder
- 210 130 - Cross-Attention Fusion Core
- 140 - Adaptive Translation Policy Engine
- 150 - Privacy-Preserving Embedding Layer
- 160 - Predictive Misunderstanding Detection Module
- 170 - Emotion-Preserving Speech Synthesis Module
- 215 180 - Expression-Aligned Visual Rendering Module
- 190 - Latency Constraint (<100 ms per Turn)

DETAILED DESCRIPTION OF THE INVENTION

The present invention will now be described in detail with reference to the accompanying drawings. The following description is provided to enable a person skilled in the art to implement
220 the invention and is not intended to limit the scope of the invention.

The invention provides a real-time multimodal adaptive communication system configured to facilitate Hindi–English cross-lingual interaction by incorporating emotional context derived from

multiple modalities. The system comprises a plurality of interconnected modules that operate in a coordinated pipeline to process input data, detect emotional state, adapt translation behavior, and generate emotion-preserving outputs.

Referring to Fig. 1, the system comprises an input interface (100) configured to receive user input in the form of speech and visual data. The input interface (100) includes a microphone for capturing audio signals and a camera for capturing facial expressions of one or more users. The captured data is forwarded to downstream processing modules for further analysis.

The speech input received through the input interface (100) is provided to a speech recognition and translation module (110). The module (110) is configured to perform automatic speech recognition to convert spoken input into textual form and further translate the recognized text from a source language to a target language, particularly between Hindi and English. The module (110) may employ neural sequence models for performing translation and may incorporate contextual cues to enhance translation accuracy.

In parallel, the system processes emotional information through a multimodal emotion recognition engine (120). The engine (120) comprises a speech encoder (120A), a facial encoder (120B), and a prosodic encoder (120C). The speech encoder (120A) extracts acoustic features from the audio signal, including spectral representations indicative of emotional content. The facial encoder (120B) processes video frames to detect facial features and infer emotional expressions of one or more individuals present in the scene. The prosodic encoder (120C) extracts temporal and pitch-related characteristics such as intonation, rhythm, and energy from the speech signal.

The outputs of the speech encoder (120A), facial encoder (120B), and prosodic encoder (120C) are provided to the multimodal emotion recognition engine (120), which generates modality-specific emotion embeddings. These embeddings are subsequently forwarded to a cross-attention fusion core (130). The fusion core (130) is configured to combine the modality-specific embeddings into a unified emotion representation using a cross-modal attention mechanism that dynamically assigns weights to each modality based on contextual relevance.

The unified emotion embedding generated by the fusion core (130) is supplied to an adaptive translation policy engine (140). The engine (140) determines a translation policy based on the emotional state and contextual parameters. The translation policy governs the manner in which the

translated output is generated and may include variations such as formal tone, casual tone, empathetic response, simplified phrasing, or clarification-oriented output.

255 The adaptive translation policy engine (140) interacts with the speech recognition and translation module (110) to influence translation output according to the selected policy. The translation output is thereby modified not only based on linguistic content but also based on detected emotional context.

260 The system further comprises a privacy-preserving embedding layer (150) configured to process the emotion embedding generated by the fusion core (130) or policy engine (140). The layer (150) applies a privacy mechanism, including noise injection, to ensure that sensitive emotional information is protected while retaining its utility for subsequent processing.

265 Additionally, the system includes a predictive misunderstanding detection module (160) configured to analyze conversational data over time. The module (160) evaluates sequential inputs to identify patterns indicative of potential misunderstanding or communication breakdown. The module (160) generates a risk score and, based on the score, may trigger corrective actions such as requesting clarification, simplifying translation output, or adjusting communication pace.

270 The translated text and processed emotional information are provided to an emotion-preserving speech synthesis module (170). The module (170) generates synthesized speech in the target language while preserving emotional characteristics such as tone, pitch, and intensity corresponding to the detected emotional state. The synthesis process ensures that the translated output retains both semantic and affective content.

275 The system further includes an expression-aligned visual rendering module (180) configured to generate a visual representation of a speaker. The module (180) renders facial expressions corresponding to the detected emotional state, thereby enhancing communication clarity in audiovisual interactions. The rendering may be performed using predefined mappings between emotional states and facial action parameters.

280 The entire processing pipeline operates under a latency constraint (190), ensuring that the system delivers real-time performance with response time maintained below a predefined threshold, such as 100 milliseconds per interaction cycle. The latency constraint (190) enables seamless conversational experience in live communication scenarios.

Referring now to Fig. 2, the figure illustrates confusion matrices corresponding to multiple classification models evaluated on a test dataset. The confusion matrices depict prediction distributions across a plurality of emotion classes for a speech-based emotion classification model, a vision-based facial emotion classification model, and a multimodal fusion-based classification model. Additionally, a confusion matrix corresponding to a translation policy classification model is illustrated, showing prediction accuracy across a plurality of policy categories. The figure demonstrates classification performance and inter-class confusion characteristics of individual and combined models.

Referring to Fig. 3, the figure illustrates a comparative performance analysis of multiple models in terms of accuracy and F1-score. The graphical representation highlights performance variations across individual modality models and a multimodal fusion model. The multimodal fusion model exhibits improved performance metrics, thereby indicating effectiveness of combining multiple modalities for emotion recognition.

Referring to Fig. 4, the figure illustrates modality attention analysis generated by the cross-attention fusion core (130). The figure includes representations of average attention weights assigned to speech, facial, and prosodic modalities, distribution of attention values across samples, and attention patterns corresponding to different emotional states. The analysis demonstrates dynamic weighting of modalities during fusion based on contextual and emotional variations.

Referring to Fig. 5, the figure illustrates performance evaluation metrics including accuracy, precision, recall, and F1-score for multiple models, along with inference time analysis. The figure further depicts latency measurements associated with individual processing components and the overall system pipeline, demonstrating compliance with real-time processing requirements under the latency constraint (190).

Referring to Fig. 6, the figure illustrates performance characteristics of the predictive misunderstanding detection module (160). The figure includes a receiver operating characteristic (ROC) curve, distribution of risk scores corresponding to conversational inputs, and probability distributions associated with different intervention strategies. The module (160) generates graded intervention outputs based on detected risk levels.

Referring to Fig. 7, the figure illustrates a multi-dimensional comparison of different models using
310 a radar chart representation. The chart compares performance metrics including accuracy, precision, recall, F1-score, and computational efficiency. The visualization highlights trade-offs between model performance and inference speed across different processing approaches.

Referring to Fig. 8, the figure illustrates experimental validation of dynamic cross-modal attention fusion performed by the cross-attention fusion core (130). The figure includes representations of
315 modality-wise attention distributions, emotion-conditioned attention variations, and adaptive weighting behavior across multiple samples. The analysis demonstrates that the fusion mechanism dynamically adjusts modality contributions based on detected emotional context.

Referring to Fig. 9, the figure illustrates performance characteristics of the adaptive translation policy engine (140). The figure includes distributions of policy selection across multiple
320 categories, mappings between detected emotional states and selected policies, confidence scores associated with policy predictions, and accuracy variations across emotion classes. The engine (140) dynamically selects translation policies based on emotional input.

Referring to Fig. 10, the figure illustrates detailed analysis of predictive misunderstanding detection. The figure includes risk score distributions, mappings between risk levels and
325 intervention strategies, temporal progression of risk across conversational sequences, and ROC performance characteristics. The analysis demonstrates the capability of the system to monitor and predict conversational breakdown.

Referring to Fig. 11, the figure illustrates performance of the end-to-end system pipeline. The figure includes latency breakdown across individual modules, throughput scalability analysis,
330 compliance with real-time constraints, and comparative latency across different processing components. The system demonstrates coordinated operation of all modules within predefined latency limits.

Referring to Fig. 12, the figure illustrates analysis of a unified hierarchical embedding space used within the system. The figure includes projections of embeddings derived from different
335 modalities, clustering characteristics of emotion representations, and hierarchical relationships between embeddings. The representation demonstrates improved separation and organization of emotion-related features.

Referring to Fig. 13, the figure illustrates detailed analysis of the adaptive translation policy engine (140). The figure includes distribution of policy predictions across categories, confidence score distributions, and variation of policy selection across different emotional states. The analysis indicates consistent and context-aware policy selection behavior.

Referring to Fig. 14, the figure illustrates multi-person facial emotion analysis performed using a vision-based detection system within the multimodal emotion recognition engine (120). The figure demonstrates detection and analysis of emotional states from multiple participants in video frames, including distribution of detected emotions, comparative accuracy, detection confidence levels, and simultaneous analysis of multiple individuals in a communication scenario.

In operation, the system receives multimodal input through the input interface (100), processes speech and emotional cues in parallel, fuses the multimodal data to determine emotional context, dynamically adapts translation behavior based on the emotional state, and generates emotion-preserving speech and visual outputs while maintaining privacy and minimizing communication errors.

The invention thus provides an integrated and adaptive communication framework that enhances cross-lingual interaction by incorporating emotional intelligence, thereby improving the accuracy, clarity, and effectiveness of multilingual communication.

Experimental Validation Results:

The proposed real-time multimodal adaptive Hindi–English communication system was experimentally validated using a multimodal dataset comprising speech, facial, and prosodic inputs. The evaluation was conducted to assess performance of individual modules and the overall system pipeline in terms of classification accuracy, fusion effectiveness, latency, and predictive capability.

The system was implemented using a deep learning framework and evaluated under controlled experimental conditions with reproducible configuration settings, including GPU-based execution and standardized training parameters.

1. Training Configuration

The system was trained using an audio-visual emotion dataset with a defined training and validation split. The configuration parameters used during training are provided below.

Table 1: Training Configuration

Parameter	Value
Framework	PyTorch 2.x (CUDA-enabled)
Device	GPU (CUDA) for experiments; CPU-compatible
Random Seed	42 (fully reproducible)
Training Data	Audio-visual emotion corpus (7 emotions)
Train/Val Split	80% / 20%
Batch Size	32
Optimizer	Adam (lr=1e-3, weight_decay=1e-4)
Scheduler	StepLR (step=5, gamma=0.5)
Early Stopping	Patience=5 epochs
Gradient Clipping	Max norm=1.0
Face Detection	YOLOv10n (Ultralytics)

2. Model Performance Summary (Training-Based Results)

The performance of individual modules was evaluated during training using validation accuracy, F1-score, and loss values.

Table 2: Model Performance Summary

Model	Parameters	Epochs	Val Accuracy	Val F1	Val Loss
Speech Emotion CNN + BiLSTM	~2.1M	15	41.96%	0.3445	1.4512
YOLOv10 + Facial Emotion Encoder	~0.5M	6 (early stop)	96.20%	0.9547	0.0312
Transformer Multimodal Fusion	~0.4M	10	98.00%	0.9782	0.0156
Adaptive Policy Engine	~0.04M	10	61.60%	0.6171	0.9967
Misunderstanding Detector	~0.3M	6 (early stop)	100.00%	1.0000	0.0001
Total System	~3.3M	—	—	—	—

The results demonstrate that the multimodal fusion model achieves the highest validation accuracy and F1-score, indicating effective integration of multiple modalities.

3. Experimental Evaluation on Test Data

The system was evaluated on 504 test samples to validate end-to-end performance of the multimodal pipeline. The confusion matrices illustrated in Fig. 2 show classification behavior across emotion categories and policy classes. The multimodal fusion model demonstrates improved classification consistency compared to individual modality models.

4. Multimodal Fusion Performance

The fusion mechanism was evaluated to analyze modality contribution and overall classification performance. As illustrated in Fig. 3 and Fig. 4, the multimodal fusion model achieves 98.0% accuracy with balanced attention distribution across speech, facial, and prosodic modalities. The attention analysis confirms that the system dynamically adjusts modality importance based on the input sample.

5. Inference Time and Latency Analysis

The inference time of individual modules and the overall system pipeline was measured to verify real-time performance.

Table 3: Inference Time Benchmarks

Model	Mean Latency (ms)
Speech CNN	20.71
YOLOv10 + Facial CNN	0.67
Multimodal Fusion	1.73
Policy Engine	0.68
Misunderstanding Detector	1.61
Total Pipeline	25.40
Throughput (batch=32)	1,260 samples/sec

The total pipeline latency of approximately 25.40 ms is below the real-time threshold, thereby enabling real-time conversational interaction.

6. Privacy Analysis

The privacy-aware embedding layer was evaluated for different privacy budget values to analyze the trade-off between privacy and utility.

Table 4: Privacy Analysis Results

Privacy Budget (ϵ)	Cosine Similarity	L2 Distortion	Privacy Level
0.1	0.02	~ 75.0	Maximum privacy (near-random)
0.5	0.08	~ 15.0	Very high privacy
1.0	0.14	~ 8.0	Strong privacy (default)
3.0	0.27	~ 4.0	Moderate privacy
5.0	0.56	~ 3.0	Balanced
10.0	0.80	~ 1.3	High utility

The results indicate that the system allows configurable privacy levels while maintaining usable emotion representations.

7. Ablation Study

An ablation study was conducted to evaluate the contribution of different modules to the overall system performance.

Table 5: Ablation Study Results

Configuration	Accuracy	Improvement	Description
Speech Only	42.0%	Baseline	Single modality (audio MFCC)
YOLOv10 Facial Only	96.2%	+54.2% over speech	Single modality (YOLOv10 + facial features)
Late Fusion (Average)	71.0%	+29.0% over speech	Simple averaging of predictions
Transformer Fusion	98.0%	+56.0% over speech	Cross-attention multimodal fusion
Fusion + Adaptive Policy	97.5%	+55.5% over speech	Full system with policy adaptation
Full System (All 7 Modules)	97.5%	+55.5% over speech	Complete Hindi-English pipeline with all modules

The results demonstrate that multimodal fusion significantly improves performance compared to single-modality approaches.

8. Contribution-wise Experimental Evidence

The experimental results further validate the key contributions of the invention. Dynamic cross-modal attention fusion achieves high classification accuracy with adaptive modality weighting. The adaptive translation policy engine demonstrates emotion-conditioned policy selection with measurable accuracy. Predictive misunderstanding detection achieves high accuracy and effective intervention prediction. The integrated pipeline operates within real-time latency constraints. The unified embedding space provides structured and separable emotion representations.

415 The experimental validation confirms that the proposed system achieves high accuracy in
multimodal emotion recognition, effective fusion of speech, facial, and prosodic information, real-
time processing capability, configurable privacy protection, and predictive detection of
conversational misunderstandings. The results demonstrate the technical effectiveness and
robustness of the proposed invention in real-time multilingual communication scenarios.

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CLAIMS

I/We Claim:

1. A computer-implemented system for real-time multilingual communication, the system comprising:

425 an input interface (100) configured to receive speech and visual data;
 a speech recognition and translation module (110) configured to convert input
speech into text and translate the text between a source language and a target language;
 a multimodal emotion recognition engine (120) comprising a speech encoder
(120A), a facial encoder (120B), and a prosodic encoder (120C), configured to generate
430 modality-specific emotion features;
 a fusion core (130) configured to combine the modality-specific emotion features
to generate a unified emotion embedding;
 an adaptive translation policy engine (140) configured to determine a translation
policy based on the unified emotion embedding and contextual parameters;
435 a privacy-preserving embedding layer (150) configured to process the unified
emotion embedding;
 a predictive misunderstanding detection module (160) configured to analyze
conversational inputs and generate a risk score;
 an emotion-preserving speech synthesis module (170) configured to generate
440 translated speech based on the translation policy and emotion embedding; and
 an expression-aligned visual rendering module (180) configured to generate a
visual representation corresponding to the emotion embedding;
wherein the system is configured to process the received input in real time to generate
translated output while preserving emotional context.

- 445 2. The system as claimed in claim 1, wherein the multimodal emotion recognition engine
(120) is configured to extract emotion features from speech signals, facial data, and
prosodic characteristics including pitch, energy, and temporal features.
3. The system as claimed in claim 1, wherein the fusion core (130) is configured to
dynamically combine the modality-specific emotion features to generate the unified
450 emotion embedding based on contextual relevance of each modality.

4. The system as claimed in claim 1, wherein the adaptive translation policy engine (140) is configured to select a translation policy from a plurality of policies including formal, casual, empathetic, simplified, and clarification-based translation.
- 455 5. The system as claimed in claim 1, wherein the emotion-preserving speech synthesis module (170) is configured to generate speech output by controlling prosodic parameters including pitch, speech rate, and intensity based on the unified emotion embedding.
6. The system as claimed in claim 1, wherein the expression-aligned visual rendering module (180) is configured to generate facial expressions corresponding to the unified emotion embedding using predefined facial action mappings.
- 460 7. The system as claimed in claim 1, wherein the privacy-preserving embedding layer (150) is configured to apply a noise-based mechanism to the unified emotion embedding to protect sensitive information while maintaining utility for downstream processing.
8. The system as claimed in claim 1, wherein the predictive misunderstanding detection module (160) is configured to analyze a sequence of conversational inputs and generate a
465 risk score indicative of a potential communication breakdown.
9. The system as claimed in claim 8, wherein the system is further configured to initiate at least one corrective action based on the generated risk score, the corrective action comprising clarification, simplification, or adjustment of communication pace.
10. A computer-implemented method for real-time multilingual communication, the method
470 comprising:
 - receiving speech and visual input through an input interface (100);
 - converting the speech input into text and translating the text between a source language and a target language using a speech recognition and translation module (110);
 - extracting emotion features from speech, facial, and prosodic inputs using a
475 multimodal emotion recognition engine (120);
 - combining the extracted emotion features to generate a unified emotion embedding using a fusion core (130);
 - determining a translation policy based on the unified emotion embedding using an adaptive translation policy engine (140);
 - 480 processing the unified emotion embedding using a privacy-preserving embedding layer (150);

analyzing conversational inputs to generate a risk score using a predictive misunderstanding detection module (160); and

485 generating translated speech and corresponding visual output using an emotion-preserving speech synthesis module (170) and an expression-aligned visual rendering module (180), respectively,

wherein the method enables real-time translation while preserving emotional context.

Dated this 19th day of March 2026

**Name and Signature of the Patent Agent
For Applicant**



**Ms. Divya Sridharan
Patent Agent(IN/PA-3292)
For Daniel & Gladys.**

490 **A REAL-TIME MULTIMODAL SYSTEM FOR ADAPTIVE EMOTION-AWARE**
HINDI-ENGLISH COMMUNICATION

ABSTRACT

500 The present invention relates to a computer-implemented system and method for real-time multilingual communication. The system comprises an input interface (100), a speech recognition and translation module (110), a multimodal emotion recognition engine (120), a fusion core (130), an adaptive translation policy engine (140), a privacy-preserving embedding layer (150), a predictive misunderstanding detection module (160), an emotion-preserving speech synthesis module (170), and an expression-aligned visual rendering module (180). The system is configured to receive speech and visual inputs, extract emotion features from multiple modalities, generate a unified emotion embedding, and adapt translation behavior based on the detected emotional context. The system further generates translated speech and corresponding visual output while preserving emotional characteristics and analyzing conversational inputs to detect potential misunderstandings. The invention enables real-time, emotion-aware and context-sensitive cross-lingual communication.

505 Fig 1

**Name and Signature of the Patent Agent
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